# Win-Worthy Idea for the Nebius x NVIDIA Global AI Hackathon: GapFinder, a "git bisect" for why your reproduction doesn't match the paper

Enter the **Coding and Agentic Engineering** track with **GapFinder**. When your re-run of a published systems/ML paper gives numbers that don't match the paper, GapFinder builds the artifact once, checkpoints it, and forks that sandbox state into parallel branches. Each branch tests one plausible *unstated* experimental assumption, and the result is an evidence-backed report of which hidden knob explains the gap. A standard coding-agent loop can't do this: there is no test suite to act as the oracle, so the only way to answer the question is controlled, forked experimentation from the same checkpoint.

# TL;DR

- **Track: Coding & Agentic Engineering, not Personal AI.** Personal AI is filling up with memory/voice assistants, and its sponsor stack has documented memory-poisoning weaknesses. The Coding track's newest primitive, Token Factory Sandboxes with Git-like fork/rollback, is still under-demonstrated outside one genre that is already crowded (bug-repair/PR agents). A research-reproduction vertical uses the same primitive without competing with them.
- **The idea: GapFinder.** Nemotron 3 Ultra reads the paper and artifact and builds a "claim ledger" (every number the paper reports) plus a ranked list of unstated assumptions. Sandboxes pays the expensive build/data-load prefix once, then forks one branch per assumption. Super and Nano run and triage the branches. The output is a Gap Report: "your throughput is 38% below Fig. 7 because the paper's runs almost certainly used X; here is the forked branch that matches within 5%." The founder story is Sayjad's live RusKey (SIGMOD 2024) reproduction.
- **Why it can win:** the pain is real and quantified. Per Collberg, Proebsting & Warren, only 32.3% of 402 code-backed ACM systems papers could be obtained and built within 30 minutes, and on CORE-Bench the best original agent (CORE-Agent with GPT-4o) scored only 21.48% on

CORE-Bench-Hard. The closest hackathon rival (RERUN) explicitly does *not* verify numbers. The academic state of the art (Artisan, ArtifactCopilot) runs single linear loops with no forking and no search over unstated parameters. And the demo produces the before/after, cost-per-explained-claim evidence that Nebius challenge winners are known for.

# Key Findings

## Step 1: Track decision (Coding & Agentic Engineering wins)

| Factor | Coding & Agentic Engineering | Personal |
|---|---|---|
| Field size | The hackathon lists 12,158 registered participants overall devpost ↗ | Same pool |
| Visible crowding | Heavy in one genre. Public repos include Arborist (tree search over sandbox states for repo repair), RepoMedic (CI repair via parallel sandbox branches), SandForge (PR agent with checkpoint backtracking), sandcoder (best-of-N from one snapshot), PQC Factory (ticket | Heavy in the saturated "ass with memory" Mnemos (mem skills, Nano/Super/U tiering), a priv voice assistan memory/notes |

| Factor | Coding & Agentic Engineering | Personal |
| --- | --- | --- |
| | to PR), ProofPatch, InfraSentinel, gulutux github +8 ↗ | Locus (self-tau skills with autc generated leas privilege polici and the "Herm Buddy" starte template GitH ↗ |
| Sponsor tech novelty | Sandboxes are in **Beta**. VM-level isolation, Git-like branching ("fork from any checkpoint, run parallel explorations, and roll back instantly"), OCI image import, per-execution CPU/memory/I/O metrics, async operations, an MCP server. github ↗ GitHub ↗ Nebius lists "Research and experimentation" as a use case itself nebius ↗ | NemoClaw (announced M 16, 2026) wrap agents like Hermes/Open in OpenShell sandboxes wit policy, networ inference rout GitHub ↗ Penligent ↗ Hermes offers persistent mei SKILL.md skill creation, 20+ messaging platforms. NVI |

| Factor | Coding & Agentic Engineering | Personal |
| --- | --- | --- |
| | | has already published a reference Hermes+Nem research-agen blueprint, so tl "obvious" shov already exists NVIDIA Devel ↗ Nousresearch ↗ |
| Hidden risk | Being seen as "yet another repair agent" | A 2026 arXiv s (2606.04329, " Untrusted Inpu Trusted Memo found Hermes an average me poisoning atta success rate o 66.67% vs 34.2 for OpenClaw (retrieval succ 64.70% vs 17.4 because persi memory is inje |

| Factor | Coding & Agentic Engineering | Personal |
| --- | --- | --- |
| | | into every sess generic assista demo invites t critique |
| Team insider fit | Very strong: a live RusKey reproduction in WSL2, long-running experiments, HANDOFF.md/RESULTS_LOG.md session hand-offs | Moderate: Age and PDPA priv work are relev but Locus alre occupies "secu self-learning s |
| Judge fit | Token Factory founding PMM, Nebius DevRel, ML/AI product heads, and NVIDIA engineers reward a product that sells Sandboxes + tiered Nemotron devpost ↗ | Good for NVID NemoClaw engineers, but directly tied to Token Factory product |

**Decision:** Coding & Agentic Engineering. The crowding there is concentrated in "fix failing tests." Everyone in that cluster uses branching to find a patch that turns tests green. Nobody visible uses branching as a **scientific instrument**, i.e. a controlled experiment where the oracle is a number printed in a PDF rather than a test suite. That is the open lane.

## Step 2: Candidate generation and the five-question filter

| Candidate | Q1 audience + pain | Q2 load-bearing tech | Q3 non-obvious | be |
|---|---|---|---|---|
| **GapFinder**: forked-sandbox search over unstated paper assumptions | ✅ Grad students, artifact-evaluation committees, and ML/systems engineers adopting papers | ✅ Breaks without fork-from-checkpoint and without a 1M-context Ultra reading paper + code | ✅ Not repair, not smoke-testing | ✅ cla ex co ex cla ru |
| RERUN-style "does the repo run?" checker | ✅ | Partial | ❌ Already built in this hackathon | ✅ |
| Personal "lab notebook" Hermes agent that watches long | ✅ | ✅ (Hermes memory/cron) | ❌ Reads as "second brain for researchers" | W |

| Candidate | Q1 audience + pain | Q2 load-bearing tech | Q3 non-obvious | be |
|---|---|---|---|---|
| experiments and writes HANDOFF.md | | | | |
| Skill/memory quarantine for Hermes (AgentIAM angle) | ✅ | ✅ | Partial (Locus is adjacent) | ✅ su be |
| PDPA-compliant telecom assistant (Bangladesh) | Medium | Weak | ✅ | W |

# Details: The Final Idea

## Chosen track and why

**Coding & Agentic Engineering.** GapFinder is literally "an agent that writes, runs, and tests code in Token Factory Sandboxes," which is the track definition verbatim.

Devpost
↗
devpost
↗
Its "tests" are the paper's own reported numbers. Personal AI would force the idea into an "assistant" framing that judges have seen many times.

## Name + one-line pitch

**GapFinder: "Your numbers don't match the paper. GapFinder forks your experiment 30 ways and tells you which unstated assumption is why."**

## Named audience and quantified pain

- **Primary audience:** PhD/MS students and research engineers reproducing systems and ML papers. This is often the first assignment in a new lab, and it is especially hard in the Global South, where compute and advisor time are scarce.
- **Secondary audience:** Artifact Evaluation (AE) committees at ACM/USENIX/IEEE venues. They award "Results Reproduced" badges, which ACM defines as main results "obtained in a subsequent study by a person or team other than the authors." ACM
↗
- **Pain, quantified:**
  - **Code often doesn't even build.** Collberg, Proebsting & Warren (University of Arizona TR, later Commun. ACM 2016) started from 601 papers and examined the 402

whose results were backed by code; for only 32.3% of these could they obtain the code and build it within 30 minutes.

- **Reproduction is hard even for frontier agents.** On CORE-Bench (270 tasks from 90 papers), the best original agent (CORE-Agent with GPT-4o) scored only 21.48% on CORE-Bench-Hard. On the Holistic Agent Leaderboard, the top CORE-Bench-Hard result was Claude Opus 4.1 at 51.1%, costing $412.42 for the run. arxiv ↗
- **Setting up research repos is its own benchmark problem.** SUPER (EMNLP 2024) was created because agents struggle to set up and execute tasks from low-profile research repositories. ACL Anthology ↗ ResearchGate ↗
- **Even when code runs, numbers drift.** Raff (NeurIPS 2019) independently replicated 162 of 255 ML papers (63.5%) without using the authors' code, counting a paper as reproduced when at least 75% of its claims held. Belz et al.'s 2021 systematic review of NLP reproducibility found just 14% of 513 original/reproduction score pairs (77) were exactly the same, often because of underspecified conditions. A meta-analysis of 142 ReScience C replication studies names missing experiment details such as hyperparameters as a cause. Awesomepapers ↗

In 50 RL papers, only 20% specified how hyperparameters were chosen. arxiv ↗

- **Paper-vs-artifact inconsistencies are common.** The Artisan agent found 20 paper–artifact inconsistencies in 13 of 23 software-engineering papers (57%), and authors confirmed 14 of 16 reported ones. arxiv ↗

The pain GapFinder targets is not "the code doesn't run" (partially addressed elsewhere). It is the **last, most expensive mile**: "it runs, but my Figure 7 is 38% off and I don't know if it's me, the hardware, or something the paper never said."

## The founder narrative (demo-video hook)

*"I'm Sayjad, from BUET's CSE ML group. For weeks I've been reproducing RusKey (SIGMOD 2024), a RocksDB fork with an RL tuner that picks the compaction policy per level. I work in WSL2 on Windows, with a coding agent, and I pass state between sessions through HANDOFF.md and RESULTS_LOG.md. Every time my numbers disagreed with the paper, I faced the same question: what did the authors do that they didn't write down? Key-value size? Block cache? Warm-up length? Compiler flags? Testing each one meant rebuilding and reloading from scratch. So we built the tool I needed."*

This has the same shape as the winning founder stories (a lawyer building for permit rejections, a physician

building Medkit, a lecturer building Maieutic). It is a lived, ongoing, specific pain, told by the person who has it.

## Why NVIDIA/Nebius tech is load-bearing

### Token Factory Sandboxes: the product *is* the fork.

- **Expensive prefix paid once.** Cloning, building a C++ RocksDB fork, and loading a multi-GB key-value dataset happen once. The resulting state becomes an immutable checkpoint. github ↗ GitHub ↗ The docs say: "Fork execution state at any checkpoint. Explore multiple solution paths in parallel, then score results and expand the best branches." nebius ↗
- **Controlled experiments.** Every hypothesis branch starts from a byte-identical state, so differences between branches can be attributed to the one knob that changed. That is the scientific core, and it only works with forkable state.
- **Built-in resource metrics** (CPU time, memory, I/O per execution) give a noise baseline for performance claims. **Instant rollback** lets bisection search walk back up the tree. GitHub ↗ nebius ↗

- **The checkpoint replaces HANDOFF.md.** Checkpoints are retained for 180 days in Beta, so a reproduction becomes a shareable, replayable state tree instead of a markdown file describing one. nebius ↗
- **OCI image import** lets an AE committee start from the authors' own Docker image. nebius ↗

**Nemotron tiering (each size has a job it is suited to):**

- **Nemotron 3 Ultra (550B total / 55B active, 1,024K-token context on Token Factory)** does the hard reasoning. Nebius +2 ↗
  It reads the whole paper plus the whole artifact (code, configs, scripts, README) in one context and produces (a) the **claim ledger** (every quantitative claim, with figure/table provenance) and (b) a **ranked list of unstated or inconsistent assumptions** (e.g., "the paper says 1 KB values but `db_bench` defaults differ", "warm-up is never mentioned"). It also adjudicates final attributions. NVIDIA positions Ultra for "long-running agents," and its GitHub guidance covers harnesses like OpenCode and OpenHands. NVIDIA Developer ↗
- **Nemotron 3 Super (120B / 12B active)** handles per-branch diagnosis: build failures, config conflicts, and deciding whether a branch's result moved toward or away from the claim. Nebius describes it as "optimized for multi-agent applications and complex reasoning

workflows." Nebius
↗
*(Context length conflicts across Nebius pages: the launch blog says "up to 1M," while the model catalog lists 256K. That is sufficient either way.)* Nebius
↗
Nebius
↗

- **Nemotron 3 Nano (30B / 3B active)** does the wide, cheap work: Wikipedia
  ↗
  triaging logs, parsing benchmark output into structured numbers, and one call per branch. Most branches get discarded, so the discarded work should be the cheapest. github
  ↗
- **Nemotron 3 Nano Omni (30B / 3B active, image input)** reads **plots**. Wikipedia
  ↗
  Many systems papers report key results only as figures. Extracting claimed values from Figure 7's bars is the "new vision capability is load-bearing" move, analogous to Wrench Board betting on schematic vision.

  **Serverless Jobs (optional, for heavy lifting):** GPU-bound ML experiments that exceed the sandbox profile can be dispatched as Serverless AI Jobs, "containerized finite workloads that start, run and complete." Nebius
  ↗
  This mirrors the Serverless Builders Challenge winner who ran seven ablation jobs in parallel (20 minutes vs.

three hours) for $320 total. Nebius ↗ NemoClaw/OpenShell/Hermes are *not* required for this track, and forcing them in would be decoration.

## What makes it non-obvious (novelty check)

| Existing work | What it does | Gap GapFinder fills |
|---|---|---|
| **RERUN** (this hackathon, TOUMO45/rerun) | Rebuilds a paper repo in a Token Factory Sandbox and gives a verdict at "SMOKE level: exit code 0 + non-trivial output." States it "does not verify that the paper's numerical results are reproduced." Sandboxes are destroyed after use, and its 20-repo | GapFinder starts where RERUN stops: numbers, checkpoints, forks |

| Existing work | What it does | Gap GapFinder fills |
| --- | --- | --- |
| | test set "has not been executed yet" github ↗ | |
| **Arborist / RepoMedic / SandForge / sandcoder** (this hackathon) | Fork sandbox states to search for a patch that makes tests pass github +4 ↗ | No test oracle exists for "match Fig. 7." The search space is experimental assumptions, not code patches |

| Existing work | What it does | Gap GapFinder fills |
| --- | --- | --- |
| **Artisan** (Baek & Pradel, 2026) | Agentic artifact evaluation. Reproduced 44/60 tables at $0.45 and 48 min per task (GPT-5.1), 3.14× better than mini-swe-agent (44 vs 14) arxiv ↗ | Single linear loop, a fresh container per judge run, no branching, no systematic search over unstated parameters. Its benchmark excludes GPU and non-deterministic artifacts, and it notes "a need for stronger guidance when there are paper–artifact inconsistencies" arxiv ↗ arxiv ↗ |
| **ArtifactCopilot** (Wu et al., 2026) | Matched human AE outcomes on 85.42% of 48 artifacts at | Scores mainly the *Functional* badge. Its "checkpoints" are status records, not |

| Existing work | What it does | Gap GapFinder fills |
| --- | --- | --- |
| | $0.091 each (v1). The current version reports 70.56% badge agreement on 60 artifacts arxiv ↗ arxiv ↗ | forkable filesystem states arxiv ↗ |
| **PaperBench / Paper2Code / SciReplicate / ResearchCodeBench / CORE-Bench / SUPER** | Benchmarks or generators for writing code from papers, or for running existing code arXiv +2 ↗ | They measure whether reproduction succeeds. None *explains why a reproduction diverges* arxiv ↗ |

**The non-obvious angle in one sentence:** everyone else asks "does it reproduce?", while GapFinder asks "*why doesn't it*, and which unstated assumption closes the gap?" Answering that requires controlled experiments

from identical states, which is Sandboxes' distinctive primitive.

## The evidence layer (what the project naturally produces)

- **Primary before/after metric: claim gap closed.** For each claim in the ledger, record |reproduced − reported| / reported on the naive run versus the best explained branch. Example of the headline form: "Median gap on RusKey's 9 headline claims: 34% → 6%."
- **Attribution rate:** the share of divergent claims for which GapFinder names a single assumption that closes ≥ X% of the gap, verified by a counterfactual branch.
- **Setup amortization:** the count of expensive-prefix runs, fork (1) vs. naive re-run (N). Arborist's public numbers show this effect is measurable: 12 setup runs vs. 86 for linear. github ↗
- **Cost per explained claim**, with Nemotron tokens per tier and sandbox CPU-seconds from built-in metrics. The honest total bill goes in the README, including failed branches, in the style of MediSimplifier and Balagan.
- **Discovered inconsistencies:** paper-vs-artifact mismatches found, and later confirmed or refuted by authors. This mirrors Artisan's "20 new errors." arXiv ↗ arxiv ↗

- **Negative-result honesty:** some gaps will be "unexplained: likely hardware." Reporting that rate openly is what makes the numbers credible to NVIDIA staff engineers.

*All figures in the examples above are illustrative targets, not measured results.*

## The single "wow" demo moment

The screen splits. On the left is the paper's Figure 7 (RusKey throughput bars); Nano Omni has already extracted the claimed values. On the right, one checkpoint ("RocksDB-fork built + 10M keys loaded") **fans out into ~24 live sandbox branches**, each labeled with a hypothesis Ultra pulled from the paper/code mismatch. Bars grow in real time. Most branches stay red and get pruned. One turns green: "*Unstated assumption: benchmarks run after a warm-up compaction phase that the paper never mentions. Gap: 38% → 4%.*" The camera cuts to Sayjad: "That was three weeks of my life. It took 11 minutes and $X."

## Scoring self-assessment (four equally weighted Stage-Two criteria)

| Criterion | Self-score | Why |
|---|---|---|
| **Technological Implementation** | 9/10 | Genuine multi-tier Nemotron routing (Ultra for ledger/adjudication, Super for diagnosis, Nano for fan-out, Nano Omni for figures). Sandboxes fork/rollback/metrics are the core mechanism, not a wrapper |
| **Design** | 8/10 | A coherent end-to-end experience: paper + repo in, claim ledger, live fork tree, Gap Report plus a shareable replayable checkpoint out. Risk: tree UIs can overwhelm, so the report must lead |
| **Potential Impact** | 9/10 | A named audience (reproducing students, AE committees), cited pain statistics, and a demonstrated fix on a real |

| Criterion | Self-score | Why |
| --- | --- | --- |
| **Quality of the Idea** | 9/10 | SIGMOD 2024 paper the team is actually reproducing<br><br>Non-obvious use of branching as a controlled-experiment instrument. Shows real understanding of AE badges, underspecification, and noise in performance measurement |

**Stage One pass:** Genuine track fit (writes, runs and tests code in Token Factory Sandboxes). Required APIs used reasonably (runtime Token Factory inference calls to NVIDIA Nemotron). Devpost
↗
GitHub
↗
Runs on Token Factory, with optional Nebius Serverless Jobs. devpost
↗

## Why NVIDIA/Nebius would showcase it

- It is the flagship story for a Beta product Nebius itself pitches for "Research and experimentation." It is also a non-repair use of fork-from-checkpoint, which differentiates Sandboxes from generic container runners.

nebius
↗

- It exercises the full Nemotron 3 ladder on one task, with a cost-per-tier story that matches the hackathon brief: "Reach for Nemotron 3 Ultra when you need serious reasoning, and let Nano or Super handle the fast, everyday calls." devpost
↗
- It speaks to the scientific community NVIDIA courts (Nemotron's own reports, SIGMOD/NeurIPS AE committees).

**Tavily bonus ($3,000):** It qualifies naturally, but as a secondary element. devpost
↗
Genuine uses: fetching the paper's GitHub issues, errata, author talks/slides, and library release dates to rank hypotheses. Arborist found models are "confidently wrong about a library exactly when its training snapshot predates the version." github
↗
Pitch Tavily as "external evidence for hypotheses," not as the core.

## Risks of being seen as generic, and how the framing avoids them

| Risk | Mitigation in the framing |
| --- | --- |
| "Another sandbox-branching repair | Never say "fix," "patch," or "PR." Say "explain the gap," "claim ledger," |

| Risk | Mitigation in the framing |
| --- | --- |
| agent" | and "controlled experiment." The oracle is the paper's number, not a test suite |
| "Paper2Code clone" | GapFinder does not write the paper's code. It *interrogates* existing artifacts |
| "RERUN duplicate" | Lead with RERUN's own disclaimer ("does not verify … numerical results") as the contrast github ↗ |
| Performance noise in microVMs makes attributions look shaky | Frame claims as *relative* (ratios between configurations, trends across branches), and report the noise band from repeated forks. Include "unexplained" as an honest outcome |
| Sandbox Beta limits (50 concurrent operations, per-project access) nebius ↗ | Keep fan-out within 50, and state the limits openly in the Feedback submission, which also targets the "Most Valuable Feedback" prize devpost ↗ |

| Risk | Mitigation in the framing |
| --- | --- |
| Overclaiming generality | Scope to systems/ML papers with public artifacts, showcased on RusKey. Present general coverage as future work |

# Runner-up Alternatives

**1. SkillQuarantine (Personal AI).** A NemoClaw/OpenShell-hosted Hermes add-on that admits a newly self-written SKILL.md or memory write into the live assistant only after replaying it in a sandboxed fork against a battery of injection probes. It then attaches an AgentIAM-style least-privilege scope. The pain is sharply quantified: a 2026 arXiv study (2606.04329) measured an average memory-poisoning attack success rate of 66.67% on Hermes vs 34.25% on OpenClaw, and "skill misevolution" in self-improving agents is an active research topic. It fits NVIDIA's NemoClaw story and the team's AgentIAM work. It loses to GapFinder because Locus already occupies the adjacent "safe self-taught skills with auto-generated least-privilege policy" niche in this hackathon, GitHub ↗ and because the founder story is less lived.

**2. ExperimentWatch (Personal AI).** An always-on Hermes agent inside NemoClaw that babysits long-running experiments (RusKey training/benchmark runs). It keeps HANDOFF.md/RESULTS_LOG.md as structured persistent memory, pings on Telegram when a run diverges, and hands context to the next coding-agent session. It matches Sayjad's pain exactly and uses Hermes cron/memory/messaging natively. Nousresearch ↗
It loses because judges will read it as "a second brain for researchers," a saturated genre. Its best feature (session continuity) is better expressed as GapFinder's replayable checkpoints.

## Caveats

- Project-gallery visibility is limited. This hackathon's Devpost gallery and forum could not be fetched directly, so crowding was inferred from public GitHub repos that self-identify as entries, and more entries will appear before Oct 30. Re-check for "reproduction" and "artifact evaluation" entries close to submission.
- Sandbox capabilities are Beta. Whether sandboxes have GPUs, and what their egress and timeout limits are, was not confirmed in the documentation reviewed. GPU-bound ML claims may need Serverless Jobs, and performance-sensitive systems claims need careful noise handling.

- The example metrics ("34% → 6%", "38% → 4%", "11 minutes") are illustrative targets for the pitch, not measurements.
- Some statistics are old (Collberg 2014–2016, Raff 2019). They are still the canonical citations, but pair them with the 2026 Artisan/ArtifactCopilot numbers to show the problem persists. ArtifactCopilot's headline figures differ between arXiv versions (85.42% on 48 artifacts in v1 vs. 70.56% on 60 now), so cite the version explicitly. arxiv ↗
arxiv ↗